

MAT 1013: Discrete Mathematics

Session 7 : Congruences, Modular Arithmetic, Modular Inverses and RSA Cryptography

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Topics

- Congruence modulo n
- Congruence classes
- Properties of congruence
- Arithmetic modulo n
- Solving linear congruences
- Multiplicative inverses modulo n
- Extended Euclidean algorithm
- Applications of modular arithmetic
- Toy example of RSA cryptography

Learning Outcomes

By the end of this handout, students should be able to:

- Define congruence modulo n .
- Convert congruence statements into divisibility statements.
- Determine whether two integers are congruent modulo n .
- Describe congruence classes modulo n .
- Prove basic properties of congruence.
- Perform addition, subtraction, multiplication, and powers modulo n .
- Use modular arithmetic to solve divisibility and remainder problems.
- Determine whether a multiplicative inverse modulo n exists.
- Use the extended Euclidean algorithm to find modular inverses.
- Solve linear congruences using modular inverses.
- Explain the basic idea of RSA cryptography using a small numerical example.

1. Modulo Arithmetic

Activity 1: Working with Remainders

Consider the numbers:

$$2, 7, 12, 17, 22, \dots$$

1. What is the remainder when each number is divided by 5?
2. What is the difference between consecutive terms?
3. In what sense are these numbers “the same” with respect to division by 5?

Activity 2: Modulo n

Show that two integers a and b have the same remainder when divided by $n \in \mathbb{N}$ if and only if their difference $a - b$ is divisible by n .

We shall write $a \equiv b \pmod{n}$ if $n \mid a - b$

Activity 3: Working with congruence classes modulo 5

Use the table below to answer the following questions.

Remainder 0 modulo 5	remainder 1 modulo 5	remainder 2 modulo 5	remainder 3 modulo 5	remainder 4 modulo 5
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$
-15	-14	-13	-12	-11
-10	-9	-8	-7	-6
-5	-4	-3	-2	-1
0	1	2	3	4
5	6	7	8	9
10	11	12	13	14
15				
$\vdots$	$\vdots$	$\vdots$	$\vdots$	$\vdots$

- We call each column a modulo 5 congruence class.
- Each class can be identified by the least non-negative remainder. For example the first column is the congruence class of 0.

1. Choose two numbers from the column congruent to 2 modulo 5. Add them. Which column does the answer belong to?
2. Choose one number from the column congruent to 1 modulo 5, and one number from the column congruent to 3 modulo 5. Add them. Which column does the answer belong to?
3. Complete the following:

$$2 + 3 \equiv \underline{\hspace{1cm}} \pmod{5}.$$

4. Choose one number from Activity 2:Working with congruence classes modulo the column congruent to 4 modulo 5, and one number from the column congruent to 1 modulo 5. Subtract the second number from the first. Which column does the answer belong to?
5. Complete the following:

$$4 - 1 \equiv \underline{\hspace{1cm}} \pmod{5}.$$

6. Choose one number from the column congruent to 2 modulo 5, and one number from the column congruent to 4 modulo 5. Multiply them. Which column does the answer belong to?
7. Complete the following:

$$2 \cdot 4 \equiv \underline{\hspace{1cm}} \pmod{5}.$$

8. Choose two numbers from the column congruent to 3 modulo 5. Multiply them. Which column does the answer belong to?

9. Complete the following:

$$3 \cdot 3 \equiv \underline{\hspace{1cm}} \pmod{5}.$$

10. Based on your observations, complete the following tables. Here, $[i]$ represents a congruence class and the answer is also a congruence class.

$+_5$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$	$\times_5$	$[0]$	$[1]$	$[2]$	$[3]$	$[4]$
$[0]$						$[0]$					
$[1]$						$[1]$					
$[2]$						$[2]$					
$[3]$						$[3]$					
$[4]$						$[4]$					

Activity 4:

Where do you get to see modulo arithmetic?

2. Congruence Modulo n

Definition : Congruence Modulo n

Let $a, b \in \mathbb{Z}$, and let $n \in \mathbb{N}$. We say that a is congruent to b modulo n , written

$$a \equiv b \pmod{n},$$

if

$$n \mid (a - b).$$

Equivalently, a and b leave the same remainder when divided by n .

Examples

- $17 \equiv 2 \pmod{5}$, because $17 - 2 = 15$ and $5 \mid 15$.
- $40 \equiv 4 \pmod{6}$, because $40 - 4 = 36$ and $6 \mid 36$.

Activity 5: Congruence Check

Determine whether each statement is true or false. Justify your answer using divisibility.

- $23 \equiv 8 \pmod{5}$
- $40 \equiv 4 \pmod{6}$
- $31 \equiv 7 \pmod{8}$
- $123 \equiv 18 \pmod{7}$
- $-3 \equiv 9 \pmod{4}$
- $100 \equiv 1 \pmod{9}$

3. Congruence Classes

Definition : Congruence Classes

The congruence class of a modulo n is

$$[a]_n = \{x \in \mathbb{Z} : x \equiv a \pmod{n}\}.$$

It consists of all integers that leave the same remainder as a when divided by n .

Examples

There are exactly five congruence classes modulo 5. Namely:

$$[0]_5, [1]_5, [2]_5, [3]_5, [4]_5.$$

For example,

$$[0]_5 = \{\dots, -10, -5, 0, 5, 10, 15, \dots\},$$

$$[1]_5 = \{\dots, -9, -4, 1, 6, 11, 16, \dots\},$$

$$[2]_5 = \{\dots, -8, -3, 2, 7, 12, 17, \dots\}.$$

Activity 6: Listing Congruence Classes

1. List all congruence classes modulo 3.
2. List all congruence classes modulo 4.
3. Which congruence class modulo 6 contains 25?
4. Which congruence class modulo 7 contains -2 ?

4. Congruence as an Equivalence Relation

A relation is an *equivalence relation* if it is (i) reflexive, (ii) symmetric, and (iii) transitive.

- **Reflexive:** $\forall a \in \mathbb{Z}, \quad a \equiv a \pmod{n}$
- **Symmetric:** $\forall a, b \in \mathbb{Z}, \quad a \equiv b \pmod{n} \implies b \equiv a \pmod{n}$
- **transitive:** $\forall a, b, c \in \mathbb{Z}, \quad a \equiv b \pmod{n} \text{ and } b \equiv c \pmod{n} \implies a \equiv c \pmod{n}$

Equivalence relations behave *like the equal sign*.

Theorem

For a fixed positive integer n , congruence modulo n is an equivalence relation on $\mathbb{Z}$.

Proof. **Reflexive.** For every $a \in \mathbb{Z}$,

$$a - a = 0.$$

Since $n \mid 0$, we have

$$a \equiv a \pmod{n}.$$

Symmetric. Suppose

$$a \equiv b \pmod{n}.$$

Then

$$n \mid (a - b).$$

Hence there exists $k \in \mathbb{Z}$ such that

$$a - b = nk.$$

Then

$$b - a = -nk = n(-k).$$

Since $-k \in \mathbb{Z}$, we have

$$n \mid (b - a).$$

Therefore

$$b \equiv a \pmod{n}.$$

Transitive. Suppose

$$a \equiv b \pmod{n}$$

and

$$b \equiv c \pmod{n}.$$

Then

$$n \mid (a - b)$$

and

$$n \mid (b - c).$$

Therefore n divides their sum:

$$(a - b) + (b - c) = a - c.$$

Hence

$$n \mid (a - c),$$

so

$$a \equiv c \pmod{n}.$$

Thus congruence modulo n is an equivalence relation. □

5. Properties of Modular Arithmetic

Theorem

Suppose

$$a \equiv b \pmod{n}$$

and

$$c \equiv d \pmod{n}.$$

Then:

1. $a + c \equiv b + d \pmod{n}$.
2. $a - c \equiv b - d \pmod{n}$.
3. $ac \equiv bd \pmod{n}$.
4. $a^k \equiv b^k \pmod{n}$ for any $k \in \mathbb{N}$.

Proof. **Proof of Addition Property**

Since $a \equiv b \pmod{n}$, we have $n \mid (a - b)$. Since $c \equiv d \pmod{n}$, we have $n \mid (c - d)$. Therefore,

$$n \mid ((a - b) + (c - d)).$$

But

$$(a - b) + (c - d) = (a + c) - (b + d).$$

Hence

$$a + c \equiv b + d \pmod{n}.$$

□

Proof of Multiplication Property

Since $a \equiv b \pmod{n}$, there exists $r \in \mathbb{Z}$ such that $a = b + nr$. Since $c \equiv d \pmod{n}$, there exists $s \in \mathbb{Z}$ such that $c = d + ns$. Then

$$ac = (b + nr)(d + ns) = bd + bns + dnr + n^2rs.$$

Thus

$$ac - bd = n(bs + dr + nrs).$$

Therefore

$$ac \equiv bd \pmod{n}.$$

Proof of Exponentiation

This now follows from the multiplicative property. □

6. Computations with Arithmetic Modulo n

When computing modulo n , we may replace numbers by their remainders.

Worked Examples

1. Lets compute $17 + 29 \pmod{7}$. Since $17 \equiv 3 \pmod{7}$ and $29 \equiv 1 \pmod{7}$, we get

$$17 + 29 \equiv 3 + 1 = 4 \pmod{7}.$$

2. To compute $23 \cdot 31 \pmod{5}$, we first note that $23 \equiv 3 \pmod{5}$ and $31 \equiv 1 \pmod{5}$. Thus, we get

$$23 \cdot 31 \equiv 3 \cdot 1 = 3 \pmod{5}.$$

Activity 7: Modular Computation

Compute:

- | | |
|---------------------------|---------------------------|
| 1. $123 + 456 \pmod{7}$ | 4. $37^2 \pmod{6}$ |
| 2. $23 \cdot 31 \pmod{5}$ | 5. $99 + 101 \pmod{11}$ |
| 3. $2^{10} \pmod{7}$ | 6. $15 \cdot 18 \pmod{8}$ |

7. Patterns in Powers

Modular arithmetic often reveals repeating patterns.

Example

Compute powers of 2 modulo 5:

$$\begin{aligned} 2^1 &\equiv 2 \pmod{5}, \\ 2^2 &\equiv 4 \pmod{5}, \\ 2^3 &\equiv 8 \equiv 3 \pmod{5}, \\ 2^4 &\equiv 16 \equiv 1 \pmod{5}. \end{aligned}$$

Then the pattern repeats:

$$2, 4, 3, 1, 2, 4, 3, 1, \dots$$

Therefore,

$$2^{100} \equiv 1 \pmod{5}.$$

Activity 8: Power Patterns

Explore the patterns of the powers to determine the following:

1. $3^{100} \pmod{7}$

3. $7^{100} \pmod{8}$

2. $2^{100} \pmod{3}$

4. $5^{2026} \pmod{6}$

8. Interesting Problems Using Modular Arithmetic

Problem 1: Squares Modulo 4

Claim. For every integer n ,

$$n^2 \equiv 0 \text{ or } 1 \pmod{4}.$$

Proof. Every integer is either even or odd.

If n is even, then $n = 2k$, so

$$n^2 = 4k^2 \equiv 0 \pmod{4}.$$

If n is odd, then $n = 2k + 1$, so

$$n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 4(k^2 + k) + 1.$$

Hence

$$n^2 \equiv 1 \pmod{4}.$$

□

Problem 2: No Square is Congruent to 2 Modulo 4

Using the previous result, no integer square can satisfy

$$n^2 \equiv 2 \pmod{4}.$$

Problem 3: Divisibility of $n^3 - n$

Claim. For every integer n ,

$$6 \mid (n^3 - n).$$

Proof. We factor:

$$n^3 - n = n(n - 1)(n + 1).$$

This is the product of three consecutive integers. Among any three consecutive integers, one is divisible by 3. Also, among any two consecutive integers, one is even. Therefore the product is divisible by both 2 and 3, and hence by 6.

But, we can also prove it using congruences.

Write $n = 6q + r$ with $q \in \mathbb{Z}$ and $r \in \{0, 1, 2, 3, 4, 5\}$. Now, $n \equiv r \pmod{6}$ and hence,

$$n^3 - n \equiv r^3 - r \pmod{6}.$$

Now, it is easy task to check for the possible values of r that $r^3 - r \equiv 0 \pmod{6}$. Thus, we have that

$$n^3 - n \equiv 0 \pmod{6}$$

and hence $n^3 - n$ is divisible by 6. $\square$

Problem 4: Form of Primes Greater than 3

Claim. Every prime number greater than 3 is of the form

$$6k \pm 1.$$

Proof. By the division algorithm, every integer n can be written in one of the forms:

$$6k, \quad 6k + 1, \quad 6k + 2, \quad 6k + 3, \quad 6k + 4, \quad 6k + 5.$$

The forms $6k$, $6k + 2$, and $6k + 4$ are even. The form $6k + 3$ is divisible by 3. Thus a prime greater than 3 must be of the form $6k + 1$ or $6k + 5 = 6(k + 1) - 1$.

Thus, every prime greater than 3 is congruent to ± 1 modulo 6. $\square$

Problem 5: Last Digits of Squares

Claim. No perfect square ends in 2, 3, 7, or 8.

The last digit of an integer is its value modulo 10. Check:

$n \pmod{10}$		0	1	2	3	4	5	6	7	8	9
$n^2 \pmod{10}$		0	1	4	9	6	5	6	9	4	1

Thus a square can only end in

$$0, 1, 4, 5, 6, \text{ or } 9.$$

9. Solving Linear Congruences

A linear congruence has the form:

$$ax + c \equiv b \pmod{n}.$$

Example 1

Solve:

$$x + 5 \equiv 2 \pmod{7}.$$

Subtract 5 from both sides:

$$x \equiv 2 - 5 \equiv -3 \equiv 4 \pmod{7}.$$

Thus:

$$x \equiv 4 \pmod{7}.$$

Example 2

Solve:

$$3x \equiv 6 \pmod{9}.$$

Trying values modulo 9, we get:

$$x \equiv 2, 5, 8 \pmod{9}.$$

This shows that division in modular arithmetic must be treated carefully, as there are multiple answers.

Activity 9: Simple Congruences

Solve:

1. $x + 7 \equiv 3 \pmod{11}$

3. $2x \equiv 4 \pmod{6}$

2. $x + 15 \equiv 2 \pmod{17}$

4. $4x \equiv 8 \pmod{12}$

10. Multiplicative Inverses Modulo n

Definition : Multiplicative Inverse Modulo n

Let $a, n \in \mathbb{Z}$ with $n > 1$. An integer b is called a multiplicative inverse of a modulo n if

$$ab \equiv 1 \pmod{n}.$$

In this case, we write

$$b \equiv a^{-1} \pmod{n}.$$

Example

Modulo 7,

$$3^{-1} \equiv 5 \pmod{7}$$

because

$$3 \cdot 5 = 15 \equiv 1 \pmod{7}.$$

Activity 10: Inverses by Inspection

Find by inspection:

1. $2^{-1} \pmod{11}$

3. $4^{-1} \pmod{11}$

2. $3^{-1} \pmod{11}$

4. $5^{-1} \pmod{12}$

11. When Does a Modular Inverse Exist?

Theorem

An integer a has a multiplicative inverse modulo n if and only if

$$\gcd(a, n) = 1.$$

Reason/Proof

If a has an inverse modulo n , then there exists b such that

$$ab \equiv 1 \pmod{n}.$$

Thus

$$ab - nk = 1$$

for some integer k . Hence 1 is an integer combination of a and n , so a and n are relatively prime.

Conversely, if $\gcd(a, n) = 1$, then the extended Euclidean algorithm gives integers s, t such that

$$as + nt = 1.$$

Reducing modulo n , we get

$$as \equiv 1 \pmod{n}.$$

Thus s is an inverse of a modulo n .

The Euclidean algorithm not only tells us whether $\gcd(a, n) = 1$, but also helps us find integers x and y such that

$$ax + ny = 1.$$

Reducing this equation modulo n , we get

$$ax \equiv 1 \pmod{n}.$$

Therefore, x is the multiplicative inverse of a modulo n .

Activity 11: Does the Inverse Exist?

Determine whether each inverse exists. When exist, determine the inverse.

1. $7^{-1} \pmod{15}$

4. $14^{-1} \pmod{35}$

2. $9^{-1} \pmod{15}$

5. $17^{-1} \pmod{43}$

3. $11^{-1} \pmod{26}$

6. $99^{-1} \pmod{100}$

12. Extended Euclidean Algorithm and Modular Inverses

The extended Euclidean algorithm allows us to find integers s, t such that

$$as + nt = \gcd(a, n).$$

When $\gcd(a, n) = 1$, this gives

$$as + nt = 1.$$

Reducing modulo n , we get

$$as \equiv 1 \pmod{n}.$$

Thus

$$s \equiv a^{-1} \pmod{n}.$$

Worked Example 1

Find:

$$3^{-1} \pmod{7}.$$

Use the Euclidean algorithm:

$$7 = 2(3) + 1.$$

Now solve for 1:

$$1 = 7 - 2(3).$$

Modulo 7:

$$1 \equiv -2(3) \pmod{7}.$$

Hence:

$$3^{-1} \equiv -2 \equiv 5 \pmod{7}.$$

Worked Example 2

Find:

$$5^{-1} \pmod{17}.$$

Euclidean algorithm:

$$17 = 3(5) + 2,$$

$$5 = 2(2) + 1.$$

Back substitute:

$$1 = 5 - 2(2).$$

But

$$2 = 17 - 3(5).$$

Therefore:

$$1 = 5 - 2(17 - 3(5)) = 7(5) - 2(17).$$

Modulo 17:

$$7(5) \equiv 1 \pmod{17}.$$

Therefore:

$$5^{-1} \equiv 7 \pmod{17}.$$

Worked Example 3

Find:

$$7^{-1} \pmod{26}.$$

Euclidean algorithm:

$$26 = 3(7) + 5,$$

$$7 = 1(5) + 2,$$

$$5 = 2(2) + 1.$$

Back substitute:

$$1 = 5 - 2(2) = 5 - 2(7 - 5) = 3(5) - 2(7).$$

Since

$$5 = 26 - 3(7),$$

we get

$$1 = 3(26 - 3(7)) - 2(7) = 3(26) - 11(7).$$

Modulo 26:

$$-11(7) \equiv 1 \pmod{26}.$$

Thus:

$$7^{-1} \equiv -11 \equiv 15 \pmod{26}.$$

Activity 12: Find Modular Inverses

Use the extended Euclidean algorithm to find:

1. $11^{-1} \pmod{31}$

3. $17^{-1} \pmod{43}$

2. $13^{-1} \pmod{29}$

4. $19^{-1} \pmod{50}$

13. Solving Linear Congruences Using Inverses

Example

Solve:

$$5x \equiv 3 \pmod{17}.$$

From the previous section:

$$5^{-1} \equiv 7 \pmod{17}.$$

Multiply both sides by 7:

$$7(5x) \equiv 7(3) \pmod{17}.$$

Thus:

$$x \equiv 21 \equiv 4 \pmod{17}.$$

Therefore:

$$x \equiv 4 \pmod{17}.$$

Activity 13: Solve Using Inverses

Solve:

1. $3x \equiv 5 \pmod{7}$

4. $13x \equiv 9 \pmod{29}$

2. $7x \equiv 2 \pmod{11}$

5. $5x \equiv 8 \pmod{12}$

3. $11x \equiv 4 \pmod{17}$

6. $13x \equiv 1 \pmod{20}$

14. RSA Cryptography: A Toy Example

Motivation

RSA is a public-key cryptosystem based on modular arithmetic, prime numbers, and modular inverses. Real RSA uses very large primes; here we use very small primes only to understand the idea.

Step 1: Choose Two Primes

Choose:

$$p = 5, \quad q = 11.$$

Compute:

$$n = pq = 55.$$

Step 2: Compute Euler's Totient

For two distinct primes p and q ,

$$\varphi(pq) = (p-1)(q-1).$$

So:

$$\varphi(55) = (5-1)(11-1) = 4 \cdot 10 = 40.$$

Step 3: Choose a Public Exponent

Choose e such that

$$\gcd(e, \varphi(n)) = 1.$$

Let:

$$e = 3.$$

Since:

$$\gcd(3, 40) = 1,$$

this choice is valid.

Step 4: Find the Private Exponent

We need d such that

$$ed \equiv 1 \pmod{\varphi(n)}.$$

That is:

$$3d \equiv 1 \pmod{40}.$$

Using inspection or the extended Euclidean algorithm:

$$3(27) = 81 \equiv 1 \pmod{40}.$$

Thus:

$$d = 27.$$

Public and Private Keys

The public key is:

$$(n, e) = (55, 3).$$

The private key is:

$$d = 27.$$

Encryption

Suppose the message is:

$$m = 7.$$

The ciphertext is:

$$c \equiv m^e \pmod{n}.$$

So:

$$c \equiv 7^3 \pmod{55}.$$

Compute:

$$7^3 = 343.$$

Since

$$343 = 55(6) + 13,$$

we get

$$c = 13.$$

Decryption

To recover the message, compute:

$$m \equiv c^d \pmod{n}.$$

Here:

$$m \equiv 13^{27} \pmod{55}.$$

A direct computation is large, so we use repeated squaring.

$$\begin{aligned} 13^2 &= 169 \equiv 4 \pmod{55}. \\ 13^4 &\equiv 4^2 = 16 \pmod{55}. \\ 13^8 &\equiv 16^2 = 256 \equiv 36 \pmod{55}. \\ 13^{16} &\equiv 36^2 = 1296 \equiv 31 \pmod{55}. \end{aligned}$$

Now:

$$27 = 16 + 8 + 2 + 1.$$

Thus:

$$13^{27} = 13^{16} 13^8 13^2 13.$$

So:

$$13^{27} \equiv 31 \cdot 36 \cdot 4 \cdot 13 \pmod{55}.$$

Compute step by step:

$$\begin{aligned} 31 \cdot 36 &= 1116 \equiv 16 \pmod{55}. \\ 16 \cdot 4 &= 64 \equiv 9 \pmod{55}. \\ 9 \cdot 13 &= 117 \equiv 7 \pmod{55}. \end{aligned}$$

Therefore:

$$13^{27} \equiv 7 \pmod{55}.$$

So the decrypted message is:

$$m = 7.$$

Important

This example uses very small primes and is not secure. Real RSA uses very large primes.

Activity 14: Build a Toy RSA System

Use:

$$p = 3, \quad q = 11.$$

1. Compute $n = pq$.
2. Compute $\varphi(n) = (p-1)(q-1)$.
3. Choose $e = 3$. Check that $\gcd(e, \varphi(n)) = 1$.
4. Find d such that

$$ed \equiv 1 \pmod{\varphi(n)}.$$

5. Encrypt the message $m = 4$, that is compute

$$c \equiv m^e \pmod{n}.$$

6. Decrypt the ciphertext by computing

$$m \equiv c^d \pmod{n}.$$

15. Exercises

A. Congruence

1. (a) Determine whether $82 \equiv 5 \pmod{7}$.
(b) Determine whether $101 \equiv 2 \pmod{9}$.
(c) Find the least nonnegative residue of 2026 modulo 13.
(d) Find the least nonnegative residue of -17 modulo 5.

2. Compute using Modular Arithmetic.

- | | |
|------------------------------|-------------------------|
| (a) $45 + 78 \pmod{11}$ | (c) $2^{20} \pmod{5}$ |
| (b) $123 \cdot 456 \pmod{7}$ | (d) $9^{100} \pmod{10}$ |

3. Find the modular inverses, if it exists

- | | |
|------------------------|-------------------------|
| (a) $4^{-1} \pmod{9}$ | (c) $12^{-1} \pmod{35}$ |
| (b) $8^{-1} \pmod{15}$ | (d) $21^{-1} \pmod{55}$ |

4. Solve the following congruences.

- | | |
|-----------------------------|------------------------------|
| (a) $4x \equiv 1 \pmod{9}$ | (c) $12x \equiv 5 \pmod{35}$ |
| (b) $8x \equiv 7 \pmod{15}$ | (d) $21x \equiv 4 \pmod{55}$ |

5. **Proof Exercises:** Write complete proofs.

- (a) Prove that congruence modulo n is an equivalence relation.
- (b) Prove that if $a \equiv b \pmod{n}$, then

$$a^k \equiv b^k \pmod{n}$$

for every positive integer k .

- (c) Prove that if

$$a \equiv b \pmod{n}$$

and

$$c \equiv d \pmod{n},$$

then

$$ac \equiv bd \pmod{n}.$$

- (d) Prove that every prime greater than 3 is congruent to 1 or 5 modulo 6.
- (e) Prove that

$$n^5 - n$$

is divisible by 5 for every integer n .

- (f) Prove that no perfect square is congruent to 2 modulo 3.

6. **Challenge Problems:**

- (a) Determine all solutions to

$$x^2 \equiv 1 \pmod{8}.$$

(b) Prove that

$$2^{2n} - 1$$

is divisible by 3 for every $n \geq 1$.

(c) Show that no integer solution exists for

$$x^2 \equiv 3 \pmod{4}.$$

(d) Find the last digit of

$$7^{2026}.$$

(e) Find the remainder when

$$3^{100} + 4^{100}$$

is divided by 7.

(f) Solve:

$$17x \equiv 5 \pmod{43}.$$

(g) Use the extended Euclidean algorithm to solve:

$$35x \equiv 1 \pmod{64}.$$

(h) Show that if p is an odd prime and

$$a^2 \equiv b^2 \pmod{p},$$

then

$$a \equiv b \pmod{p}$$

or

$$a \equiv -b \pmod{p}.$$

7. Independent Investigation

- (a) Why does RSA work?
- (b) What is Euler's totient function?
- (c) Why must p and q be prime in RSA?
- (d) What is the Chinese Remainder Theorem?
- (e) How large are primes used in modern RSA?
- (f) Why is it easy to multiply two large primes but difficult to factor their product?